UNIVERSITY OF EDUCATION, WINNEBA
DEPARTMENT OF ICT EDUCATION
REFLECTIVE PRACTICE

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COVERAGE

INSTITUTION: Baidoo Bonsoe Senior High Technical School
DEPARTMENT: ELECTIVE ICT
CLASS: SHS 2
CLASS SIZE: 48
TOPIC: Analyzing Numerical Data Using Spreadsheet Software
SUB-TOPIC: Basic Formulae and Functions (SUM, AVERAGE, MIN, MAX).
DURATION: 120 minutes
DATE: Week Ending 20th January, 2026

INSTRUCTIONAL OBJECTIVES

By the end of the two lessons, students were able to:

1. Write and apply a basic formula (e.g. =A1+B1) in Microsoft Excel.
2. Use the SUM, AVERAGE, MIN and MAX functions to summarize a set of marks.

INTRODUCTION

Basic Formulae and Functions

I opened with a quick brainstorm: why is it better to use a formula than to add numbers by hand? A few students who help with family shops or church accounts already had a rough answer: it is faster, and you do not lose count partway through. I let that carry the lesson into the bigger idea of automation, that a formula is more accurate and quicker than doing the same sum by hand, and less likely to go wrong from a careless slip.

Students reopened the workbook they had started the previous week. I projected the formula bar and worked through a simple formula, =A1+B1, before moving to =SUM(B2:B10), keeping the marks dataset on screen the whole time so students could watch the number in the cell change as I edited the formula.

Activity 1 (Formulae and SUM): Students worked in their groups to apply SUM to a set of class marks I had prepared, then tried MIN and MAX to find the highest and lowest scores. A few who spotted the pattern quickly went ahead and tested the same formulas on a different column without being asked.

Activity 2 (AutoFill and a Family Budget): I demonstrated AutoFill, dragging a formula down a column instead of retyping it, which drew a genuine reaction from a couple of students who had been retyping formulas manually. In their groups, students then built a short family budget sheet and applied SUM and

AVERAGE to it. This was the point where the exercise stopped feeling like a routine lab task. Several groups started adding their own budget items, transport, food, school fees, instead of the ones I had suggested.

CLOSURE

I ended both lessons by returning to the essential questions rather than just asking whether there were questions. How does SUM promote responsibility in data handling? How can a computer decide pass or fail automatically, and what could go wrong if the formula is set up incorrectly? Framing the recap this way pushed the conversation past "SUM adds numbers" toward why getting the formula right actually matters when the output affects someone's grade.

STRENGTHS

1. Building on prior knowledge: reopening the previous week's workbook meant students were not starting cold. They recognized the interface and moved into the new functions quickly.
2. Demonstration before practice: working through =A1+B1 and =SUM() live, with the marks dataset visible, gave students something concrete to copy rather than an abstract explanation of syntax.
3. Relevance: the family budget task made the numbers personal. Students who were usually quiet during a routine lab task got noticeably more talkative once the sheet was about their own money.
4. Growing confidence: by the end of the Lesson, most groups were applying SUM, AVERAGE, MIN and MAX correctly with little prompting, and several were visibly proud to be handling a tool they associate with real office work.
5. Students who got problems right were clearly pleased with themselves, and some began helping neighboring groups without being asked.

CHALLENGES

1. With earlier weeks, not every student had their own laptop, so some groups worked two or three to a machine and I also took teachers laptop for them to use, which cut into the hands-on time of whoever was not at the keyboard.
2. Pace mismatch: a handful of students who had grasped SUM and AutoFill quickly in Lesson 1 were ready to move faster into Lesson 2's IF logic, while others were still consolidating the basics.

INNOVATIVE SOLUTIONS

1. I structured the budget and grading tasks so the student not at the keyboard had a specific job, reading out the next mark, checking the formula against the printed steps, or recording the group's results in their notebook, so device-sharing did not mean some students were sitting out.
2. Students who finished early were pointed to a harder version of the same task, a larger dataset which kept them engaged without splitting the lesson into a separate track, and often made them the first ones I paired with a struggling group.

LESSONS AND INSIGHTS GAINED

1. Every function landed better once students typed it themselves against real data than when I only explained it. The budget sheet made this obvious, engagement visibly rose the moment the numbers became theirs.
2. Peer support is a resource, not a distraction. Pairing a fast group with a stuck one solved problems faster than me circulating alone, and gave the fast group a reason to consolidate what they already knew.

3. Giving the non-typing student in a shared-device group a defined job kept the lesson fair instead of quietly leaving one student behind.
4. Asking why SUM promotes responsible data handling got more out of students than asking them to restate the function's syntax.

CONCLUSION

Reflecting on this week's lessons was useful mainly because it forced me to separate what looked like a small technical hiccup, a misplaced bracket, from what it was actually revealing. That functions needed a slower, more structured introduction than a straight demonstration. The pattern that keeps repeating across my lessons showed up here again too. Students learn a tool fastest when it is doing something that matters to them, and the biggest barriers are almost always practical, a shared laptop, a mistyped formula, rather than a lack of interest. I am carrying the printed reference card idea and the "give the non-typist a job" habit into the rest of the term.